MIL-D-60014B (AR)
15 April 1975
Superseding
MIL-D-60014A (MU)
1 May 1965 and
MIL-D-60013A (MU)
1 May 1965
(see 6.6)

MILITARY SPECIFICATION

DETONATOR, STAB M47, PARTS FOR AND LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1. This specification covers the parts for, and the loading, assembling and packing for one type of detonator designated as Detonator, Stab M47.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Packing, Methods of
*MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-1169- Packaging, Packing and Marking for Shipment of Inert Ammunition Components

FSC: 1375

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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*(For meaning of asterisk (*) see paragraph 6.5).

DRAWINGS

U.S. ARMY PICATINNY ARSENAL

PRODUCT AND PACKING DRAWINGS

9297872 - Disc, Detorator
9297871 - Disc, Detonator Closing
9297870 - Cup, Detonator
8837989 - Detonator Assembly, M47
*9258182 - Box, Set-up, Packing Ammunition, Inner for Small Explosive Components
*9258183 - Box, Set-up, Packing Ammunition, Outer for Small Explosive Components
*9258184 - Box, Packing, Ammunition for Small Explosive Components

*INSPECTION EQUIPMENT DRAWINGS

9200390 - Firing Assembly
8816367 - Firing Pin Guide
9200012 - Firing Pin
8827934 - Steel Ball
8827863 - Sleeve
8816368 - Lead Disc
9200602 - Bushing

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

*2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Designation E300 - Industrial Chemicals, Recommended Practice for Sampling

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Penn. 19103).

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with applicable drawings and specifications.

*3.2 Parts. - The parts shall comply with all requirements specified on Drawings (dwg) 9297872, 9297871 and 9297870 as applicable, and with all requirements specified in applicable specifications.

3.3 Detonator. - The detonators shall comply with all requirements specified on Drawing (dwg) 8837989), and with all requirements specified in applicable specifications.

3.4 Moisture content of lead azide. - The moisture content of the lead azide used in loading the detonators shall not exceed 0.50 percent.

3.5 Moisture content of primer mixture. - The moisture content of the primer mixture used in loading the detonators shall not exceed 0.30 percent.

3.6 Functioning.

*3.6.1 Output. - The detonators shall function on the first attempt and shall cause perforation of the lead disc to a diameter of not less than 0.156 inch (in.).

3.6.2 Booster initiation. - The detonators shall function and cause the booster to perforate two lead discs.

3.6.3 Waterproofness. - The detonators shall be waterproof as evidenced by normal functioning after being immersed under a minimum (min) of 2 in. of water for a min of 48 hours.

3.7 Workmanship. - All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner, and all manufacturing, processing, and assembly operations shall be correctly performed. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required),

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corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part or assembly nor shall they be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. Required packing shall be dry.

3.8 First article inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

#4.2 Classification of inspections. - The following types of inspection shall be conducted on this item.

a. First Article Inspection

b. Quality Conformance inspection

4.3 First article inspection.

#4.3.1 Submission. - The contractor shall submit a first article sample (see 6.1b) as designated by the Contracting Officer. The first article sample shall consist of twenty-five (25) complete sets of parts (i.e. twenty-five (25) of each and every component) and three hundred and sixty-five (365) loaded assemblies. The first article sample shall be evaluated in accordance with the provisions of 4.3.2 and 4.3.3.

#4.3.2 Inspections to be performed. - See MIL-A-48078, paragraphs 4.4.2.1 through 4.4.2.3 and Table I specified herein.

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TABLE I

FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60014B (AR)

PARAGRAPH #	TITLE: Detonator, Stab M47, Metal Parts		SHEET 1 OF 1		DRAWING NUMBER: See Below NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Metal Parts (Dwgs 9297872, 9297871 and 9297870) Examination for defects	25(a)		3.2	4.4.2.1 to 4.4.2.3
	Detonator Assemblies Dwg. 8837989 Examination for defects	50(a1)		3.3	4.4.2.5
	Testing Functioning	315		3.3	4.4.3.5.1
	Booster initiation	25		3.3	4.4.3.5.2
	Waterproofness	25		3.3	4.4.3.5.3
Note	(a1) Randomly selected from 365 items prior to testing				

NOTES: (a) 25 each and every component.

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4.3.3 Rejection. - (see MIL-A-48078).

4.4 Quality conformance inspection.

*4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of detonators shall contain:

a. Parts of one lot interfix number from one manufacturer.

b. Lead azide of the same type from not more than two lots produced by the same manufacturer in accordance with the same specification and same specification revision and which have subsequently been subjected to uniformly controlled drying and operating conditions at the loading plant.

c. Primer mixture from not more than one lot from one manufacturer.

d. RDX from not more than one lot from one manufacturer.

*4.4.2 Examination. - (See MIL-A-48078). Unless otherwise specified in the classification of defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.[1] Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60014B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
#4.4.2.1	Disc, Detonator				9297872
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major B					
131	Diameter		0.40%	3.2	Gage
132	Thickness		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60014B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
#4.4.2.2	Disc, Detonator Closing				9297871
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major B					
131	Diameter		0.40%	3.2	Gage
132	Thickness		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS MIL-D-60014B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.3	Cup, Detonator				9297870
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Cup split or cracked		100.0%	3.2	Visual
Major B					
131	Outside diameter, max		0.40%	3.2	Gage
132	Diameter of flash hole, min		0.40%	3.2	Gage
133	Concentricity of flash hole with outside diameter		0.40%	3.2	Gage
134	Thickness through bottom, min		0.40%	3.2	Gage
135	Total length		0.40%	3.2	Gage
Minor					
201	Inside diameter, min		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60014B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.4	Assembly, Prior to Loading and Closing				8837989
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Internal coating incomplete or inadequate		100.0%	3.3	Visual
Major A&B	None defined				
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60014B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.5	Detonator Assembly				8837989
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Cup split or cracked		100.0%	3.3	Visual
Major B					
131	Total length		0.40%	3.3	Gage
132	Diameter		0.40%	3.3	Gage
133	Disc loose		0.40%	3.3	Manual
134	Color of disc improper		0.40%	3.3	Visual
135	Crimp not full 360 degrees		0.40%	3.3	Visual
136	Disc punctured or torn		0.40%	3.3	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60014B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.6	Box, Set-up, Inner, After Loading and Prior to Closing Slide Cover				9258182
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Filler and cushion assembly missing		100.0%	3.3	Visual
2	Filler and cushion assembly improperly assembled		100.0%	3.3	Visual
3	Stiffner missing		100.0%	3.3	Visual
4	Slide cover assembled reversed		100.0%	3.3	Visual
Major A					
101	Release paper missing		0.40%	3.3	Visual
Minor					
201	Marking missing, incomplete or illegible		0.65%	3.3	Visual
202	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS MIL-D-60014B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
#4.4.2.7	Box, Set-up, Outer, with Inner Set-Up Boxes Assembled and Prior to Assembling Top				9258183
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Red surface on inner set-up boxes do not alternate on same side of outer set-up boxes		100.0%	3.3	Visual
2	Stiffner not facing one direction		100.0%	3.3	Visual
Major A					
101	Inner box does not fit properly in outer box		0.40%	3.3	Visual
102	Fillers missing		0.40%	3.3	Visual
Major B					
131	Label not securely applied to box		0.40%	3.3	Visual
Minor					
201	Marking missing, incomplete or illegible		0.65%	3.3	Visual
202	Box improperly taped		0.65%	3.3	Visual
203	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60014B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
*4.4.2.8	Sealed Packed Barrier Bag, Prior to Packing in Box	1 OF 1	9258183
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Waterproofness of barrier bag destroyed by rupture, puncture or separation of heat seal		0.40%	3.3	Visual
102	Evidence of crushed box		0.40%	3.3	Visual
NOTE:	Package should be opened if there is evidence of crushing				
Minor					
201	Heat seal of barrier bag located so that one or more subsequent seals cannot be applied		0.65%	3.3	Visual
202	Marking of bag misleading or unidentifiable		0.65%	3.3	Visual
203	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60014B (AR)

PARAGRAPH	TITLE			SHEET 1 OF 1	DRAWING NUMBER
#4.4.2.9	Box, Packing, Ammunition Prior to Assembling Cover and Strapping				9258184
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
131	Bags containing set-up boxes damaged		0.40%	3.3	Visual
132	Cushioning material incomplete or missing from around outer set-up boxes		0.40%	3.3	Visual
133	Marking missing, misleading or unidentifiable		0.40%	3.3	Visual
134	Bottom cushioning missing of incorrect thickness		0.40%	3.3	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60014B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9258184
#4.4.2.10	Box, Packing, Ammunition				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Strapping missing, broken, loose, or improperly fastened		0.40%	3.3	Visual/Manual
102	Box damaged		0.40%	3.3	Visual
103	Marking missing, incomplete or illegible		0.40%	3.3	Visual
Minor					
201	Wooden box cover nailed to box		0.65%	3.3	Visual
202	Metallic Seal missing		0.65%	3.3	Visual
203	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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4.4.3 Testing.

*4.4.3.1 Chemical Sampling. - A representative sample shall be randomly selected for applicable tests of 4.4.3. The procedure described in ASTM E300 shall be used to withdraw the samples. The applicable tests of 4.5 shall be performed using prescribed analytical procedure for replicate determinations given in standard analytical textbooks.

4.4.3.2 Moisture content of lead azide (see 3.4) - Major A defect. - The moisture content of the lead azide shall be determined for each lot at the time of loading. A sample of 2 grams shall be selected and tested as specified in 4.5.1. If the moisture content is in excess of the percentage specified and loading has not begun, the lot of lead azide shall be rejected until proper corrective action has been accomplished as verified by performing the moisture content test specified in 4.5.1. If loading has begun, action on the unloaded lead azide shall be as specified above and all detonators loaded with the questionable lead azide shall be rejected.

4.4.3.3 Binder-lubricant content of RDX (see dwg 8837989) - Major A defect. - The total binder-lubricant, calcium resinate and graphite content of the RDX shall be determined for each lot at the time of loading. A sample of 2 ounces of the RDX and binder-lubricant mixture shall be selected and tested as specified in 4.5.2. If the total binder-lubricant, calcium resinate or graphite is in excess of the percentages specified on the applicable drawing and if loading has not begun, the lot of RDX shall be rejected until proper corrective action has been accomplished as verified by performing the test for binder-lubricant content as specified in 4.5.2. If loading has begun, action on the unloaded RDX shall be as specified above and all detonators loaded with the questionable RDX shall be rejected.

4.4.3.4 Primer mixture.

4.4.3.4.1 Moisture content (see 3.5) - Major A defect. - The moisture content of the primer mixture shall be determined for each lot at the time of loading. A sample of 0.7 gram shall be selected and tested as specified in 4.5.3. If the moisture content is in excess of the percentage specified and if loading has not begun, the lot of primer mixture shall be rejected until proper corrective action has been accomplished as verified by performing the moisture content specified in

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4.5.3. If loading has begun, action on the unloaded primer mixture shall be as specified above and all detonators loaded with the questionable primer mixture shall be rejected.

4.4.3.4.2 Composition (see dwg 8837989) - Major A defect. - The composition of the primer mixture shall be determined for each lot at the time of loading. A sample of 0.5 gram shall be selected and tested as specified in 4.5.4. If the composition of the primer mixture fails to conform to the requirements specified on the applicable drawing and if loading has not begun, the lot of primer mixture shall be rejected. If loading has begun, detonators loaded with the questionable primer mixture shall be rejected.

4.4.3.5 Functioning.

*4.4.3.5.1 Output (see 3.6.1) - Major A defect. - A sample of 315 detonators shall be selected at random from each lot. If 3 or more detonators fail to comply with the requirements specified, the lot shall be rejected. If one or two defectives are found, a second sample of 315 detonators shall be selected at random and tested. If in the combined first and second sample a total of 4 or more defectives are found the lot shall be rejected. The test shall be performed as specified in 4.5.5.1 using test equipment as specified in 4.4.4.

4.4.3.5.2 Booster initiation (see 3.6.2) - Major A defect. - A sample of 25 detonators shall be selected at random from each lot. If one or more detonators fail to comply with the requirements specified, the lot shall be rejected. The test shall be performed as specified in 4.5.5.2 using test equipment as specified in 4.4.4.

4.4.3.5.3 Waterproofness (see 3.6.3) - Major A defect. - A sample of 25 detonators shall be selected at random from each lot. If one or more detonators fail to comply with the requirement specified, the lot shall be rejected. The test shall be performed as specified in 4.5.6 using test equipment as specified in 4.4.4.

4.4.3.6 Heat seal test of sealed bag (see dwg 9258183) - Major A defect. - Sampling, acceptance criteria and test procedures for this test shall be as specified in MIL-P-116.

4.4.4 Inspection equipment. - The inspection equipment required to perform the inspections and tests prescribed in

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this specification is identified in the 'Paragraph Reference/ Inspection Method' column in the tables starting with paragraph 4.4.2.1, and the test method paragraphs (see 4.5). The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and section 6.2 herein.

4.5 Test methods and procedures.

*4.5.1 Determination of moisture content of lead azide. - The moisture content of the lead azide shall be determined in accordance with Method 101.5 of MIL-STD-650 except that a two (2) gram sample shall be tested and the dish shall be covered with a sheet of polytetrafluoroethylene to form a close fit.

4.5.2 Binder-lubricant content of RDX. - The calcium resinate, graphite, and total binder-lubricant content of the RDX to be used in the detonator shall be determined as follows:

*4.5.2.1 Calcium resinate. - An accurately weighed portion of 5 grams of the sample shall be transferred to a tared medium porosity sintered-glass crucible. The sample shall be extracted with three 20 milliliter (ml) portions of chloroform which have been previously saturated with RDX, maintaining each portion in contact with the residue for approximately 5 minutes before applying suction. Any lumps which may persist shall be broken up by stirring occasionally and the sides of the crucible washed down with each 20 ml portion of chloroform to insure complete removal of the calcium resinate. The residue shall be aspirated until the odor of chloroform is no longer detectable; the crucible and contents shall be placed in a drying oven at 95-105 degrees C for 1/2 hour, cooled in a desiccator and weighed. The above extraction shall be repeated, drying, and weighing with individual 20 ml portions of chloroform until the loss in weight on subsequent extractions totals less than 0.003 grams. The loss in weight of the crucible plus contents shall be calculated to percent calcium resinate in the sample on a moisture-free basis as follows:

$$\text{Percentage of Calcium Resinate} = \frac{100A}{W}$$

where:

A = loss in weight of the crucible plus contents.
W = weight of sample on moisture free basis.

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*4.5.2.2 Graphite. - The crucible and contents remaining after the removal of calcium resinate shall be washed with a minimum of three 20 ml portions of acetone, each portion maintained in contact with the residue for approximately 1 minute with continuous stirring before applying suction, until the subsequent test for RDX in the acetone washings is negative. The crucible and contents shall be washed with a minimum of 3 ml portions of acetone, each portion maintained in contact with the residue for approximately 1 minute with continuous stirring before applying suction, until the following qualitative test for RDX in the acetone washings is negative. The acetone filtrate shall be transferred to a 25 ml beaker, evaporated to dryness on a steam bath and several drops of a solution of chromotropic acid added in 36N sulfuric acid. The formation of a reddish-brown color, as compared to a pink color obtained by a blank prepared in an identical manner indicates the presence of RDX. The residue shall be aspirated until the odor of acetone is no longer detectable; the crucible and contents shall be placed in a drying oven at 95-105 degrees C for 1/2 hour, cooled in a desiccator and weighed. The weight of the residue in the crucible shall be calculated to the percentage of graphite in the sample on a moisture free basis as follows:

$$\text{Percentage of Graphite} = \frac{100A}{W}$$

where:

A = increase in weight of filtering crucible.
W = weight of sample on a moisture-free basis.

*4.5.2.3 Total binder-lubricant. - The percentage of total binder-lubricant in the sample shall be calculated by adding the percentage of calcium resinate and graphite found.

4.5.3 Moisture content of primer mixture. - The specified sample shall be accurately weighed to the nearest milligram and dried to a constant weight at 60 degrees C plus or minus 5 degrees C, or in a desiccator containing anhydrous calcium chloride. The loss in weight shall be calculated as the percentage of moisture.

4.5.4 Composition of primer mixture. - The composition of the primer mixture shall be determined by performing the following tests in the sequence specified:

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*4.5.4.1 Barium nitrate. - An accurately weighed portion of approximately 0.5 gram of primer mixture shall be transferred to a tared medium-porosity, fritted glass crucible of 20 ml capacity. Three ml of lead azide saturated distilled water shall be added at a temperature of 50 degrees C plus or minus 2 degrees C, agitated for 1 minute, and, if necessary, lumps shall be broken up with a rubber policeman attached to a glass rod. Suction shall be applied. This procedure shall be repeated five times, making a total of six, 3-ml extractions. The sample in the filtering crucible shall be rinsed three times with alcohol, dried in an oven maintained at 60 degrees C plus or minus 5 degrees C for 30 minutes, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of barium nitrate in the sample on a dry basis as follows:

$$\text{Barium nitrate, percent} = \frac{(A-B)\ 100}{W(1-.01M)}$$

where:

A = weight of crucible and sample.
B = weight of the crucible and sample after extractions with lead azide saturated water.
M = moisture content of sample obtained from 4.5.3 in percent
W = weight of sample.

4.5.4.2 Lead styphnate.

*4.5.4.2.1 Basic lead styphnate (when applicable). - The residue obtained in the determination of barium nitrate shall be extracted with six 5 ml portions of saturated ammonium acetate solution at about 25 degrees C, (hot saturated ammonium acetate shall not be used as it dissolves antimony sulfide to some extent) agitating each portion for about 30 seconds and allowing each portion to remain in contact with the residue for approximately 2 minutes. The residue shall be washed with water until the filtrate coming through the crucible appears colorless. The filtrate and washings shall be quantitatively transferred to a 250 ml calibrated volumetric flask, the beaker washed thoroughly with distilled water and diluted to the mark. Using a calibrated pipette, a 2 ml aliquot of this solution shall be transferred to a 50 ml calibrated volumetric flask and diluted to the mark with distilled water. A portion of this solution shall be trans-

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ferred to a corex glass spectrophotometric cell having a width of 1 centimeter (cm) and its absorbance density determined at a wave length of 410 millimicrons. As reference, a similar cell containing ammonium acetate solution ofthe same concentration as the sample solution shall be used. Since the cell which holds the basic lead styphnate is not identical with the reference cell the abosrbance obtained as described above shall be corrected for the difference in the amount of light which the two cells scatter and absorb. To do this, both cells shall be filled with ammonium acetate solution and the absorbance of the cell which originally contained the basic lead styphnate solution shall be measured at a wave length of 410 millimicrons. The percentage of basic lead styphnate in the sample shall be calculated on a dry basis as follows:

$$\text{Basic lead styphnate, percent} = \frac{F\,(A-B)}{W(1-.01M)D}$$

where:

A = absorbance of basic lead styphnate solution.
B = absorbance of ammonium acetate solution.
D = width of corex cell, cm.
M = moisture content of sample obtained from 4.5.3 in percent.
W = weight of sample.
*F = spectrophotometric factor. To determine the spectrophometric factor (F) weigh a 0.5 gm sample of pure lead styphnate. Transfer the lead styphnate quantitatively with 50 to 100 ml of distilled water to a 500 ml volumetric flask, add 30 ml of saturated ammonium acetate solution to the solution in the flask and dilute to mark with distilled water. Using aliquots of the solution prepare five solutions of lead styphnate ranging in concentration from .5 to 2.5 mg per 100 ml. Determine the absorbance of each solution at 410 millimicrons. Divide concentrations of lead styphnate for each of the solutions by its corresponding absorbance value. The average of the five quotients thus obtained shall be used as the spectrophometric factor (F) for the lead styphnate.

4.5.4.2.2 Determination of normal lead styphnate (when applicable). - Extract the residue obtained in the barium nitrate determination with sufficient 5 ml portions of ammonium acetate solution (saturated) at about 25°C. until the washings are colorless. Agitate each portion for about 1 minute, then allow

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each portion to remain in contact with the residue for one additional minute. Finally wash the residue with water, saturated with antimony sulfide, until the washings come through clear. Reserve the crucible and residue for the tetracene determination. Transfer the filtrate and washings quantitatively to a 250 ml calibrated volumetric flask, washing the beaker thoroughly with distilled water and dilute to the mark with distilled water. Using a calibrated pipette, transfer a 2 ml aliquot of this solution to a second 100 ml calibrated volumetric flask and dilute to the mark with distilled water. Transfer a portion of this solution to a corex glass spectrophotometric cell having a width of 1 cm and determine its absorbance at a wave length of 410 millimicrons. Use as reference, a similar cell containing ammonium acetate solution of the same concentration as the sample in the final dilution. Since the cell which holds the normal lead styphnate is not identical with the reference cell, correct the absorbance obtained as described above for the difference in the amount of light which the two cells scatter and absorb. To do this, fill both cells with ammonium acetate solution, and measure at a wave length of 410 millimicrons the absorbance of the cell which originally contained the normal lead styphnate solution. Calculate the percentage of normal lead styphnate in the sample on a dry basis as follows:

$$\text{Normal lead styphnate, percent} = \frac{F\,(A-B)}{WD}$$

where:

A = absorbance of normal lead styphnate solution.
B = absorbance of ammonium acetate solution.
D = width of corex cell, cm.
W = weight of the dried sample.
F = spectrophotometric factor. To determine the spectrophometric factor (F) weigh a 0.5 gm sample of pure lead styphnate. Transfer the lead styphnate quantitatively with 50 to 100 ml of distilled water to a 500 ml volumetric flask, add 30 ml of saturated ammonium acetate solution to the solution in the flask and dilute to mark with distilled water. Using aliquots of the solution prepare five solutions of lead styphnate ranging in concentration from .5 to 2.5 mg per 100 ml. Determine the absorbance of each solution at 410 millimicrons. Divide concentrations of lead styphnate for each of the solutions by its corresponding absorbance value. The average of the five quotients thus obtained shall be used as the spectrophometric factor (F) for the lead styphnate.

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4.5.4.3 Tetracene. - The wet residue obtained in the determination of lead styphnate shall be washed three times with alcohol, dried in an oven maintained at 60 degrees C plus or minus 5 degrees C for 30 minutes, cooled in a desiccator and weighed. The dried residue in the crucible shall be transferred to a 125 ml beaker with a stream of water, 25 ml of water shall be added, and the contents of the beaker boiled on a hot plate for 5 minutes. The contents of the beaker shall be poured into the same crucible, and the residue in the crucible washed three times with boiling water, and finally with alcohol. The crucible and contents shall be dried in an oven maintained at 60 degrees C plus or minus 5 degrees C for 30 minutes, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of tetracene in the sample on a dry basis as follows:

$$\text{Tetracene, percent} = \frac{(C-D)\ 100}{W(1-.01M)}$$

where:

C = weight of crucible and residue after extraction with saturated ammonium acetate solution.
D = weight of crucible and residue after treating with hot water.
W = weight of the sample.
M = moisture content of sample obtained from 4.5.3 in percent.

4.5.4.4 Antimony sulfide. - The weight of the residue shall be calculated from the determination of tetracene to percentage antimony sulfide in the sample on a dry basis as follows:

$$\text{Antimony sulfide, percent} = \frac{(D-E)}{W(1-.01M}$$

where:

D = weight of crucible and sample after treating with hot water.
E = weight of crucible.
W = weight of sample.
M = moisture content of sample obtained in 4.5.3 in percent.

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4.5.4.5 Lead azide. - The lead azide content in the sample of a dry basis, shall be determined by subtracting from 100 the combined percentages of barium nitrate, basic lead styphnate, tetracene and antimony sulfide.

4.5.5 Functioning.

*4.5.5.1 Output. - The detonator shall be assemblied to the applicable test fixture specified in 4.4.4, so that the firing pin will strike the sensitive end of the detonator. A steel ball 1-3/6 inch in diameter and weighing 3.95 ounces maximum, shall be dropped on the firing pin, to function the detonator, from a height of 3 inches, maximum. The height of drop shall be measured as the height from the top of the firing pin to the bottom of the ball. A detonator that fails to fire or fails to create a hole in the lead disc of the required size shall be classed defective. Methods of inspection shall be in accordance with dwgs 9200390, 8816367, 9200012, 8827934, 8827863, 8816368 and 9200602.

4.5.5.2 Booster initiation. - The detonator shall be assembled with an M123 booster and functioned as in 4.5.5.1 except that two lead discs shall be used and the drop height shall be 15 inches. A detonator that fails to initiate the M123 Booster sufficiently and create a hole in the two lead discs shall be classed defective.

4.5.6 Waterproofness. - The detonator shall be submerged in water maintained at a temperature 70 degrees F plus or minus 10 degrees at the required depth for the time specified and immediately upon removal from water subjected to the sensitivity test specified in 4.5.5.1. A detonator that fails to fire or fails to create a hole in the lead disc of the required size shall be classed defective.

4.5.7 Heat seal of sealed bag. - The adequacy of the seal on the bag shall be determined in accordance with MIL-P-116.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

*5.1.1 Level C. - The detonators shall be packaged and marked in accordance with dwgs 9258182 and 9258183.

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*5.1.2 Level C. - The detonator parts shall be packaged in accordance with MIL-STD-1169 requirements.

5.2 Packing.

*5.2.1 Level C. - The detonators shall be packed and marked as specified on dwg. 9258184.

*5.2.2 Level C. - The detonator parts shall be packed in accordance with MIL-STD-1169 requirements.

*5.3 Marking. - Detonator containers and bags shall be marked in accordance with instructions contained in dwgs 9258182, 9258183, and 9258184. Detonator parts containers shall be marked in accordance with MIL-STD-1169 requirements.

6. NOTES

6.1 Ordering data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Provisions for submission of first article samples.
c. Provisions of MIL-A-48078.

6.2 Submission of inspection equipment designs for approval. - (See MIL-A-48078). Submit equipment designs as required, to Commander: Picatinny Arsenal, ATTN: SARPA-QA-T Dover, New Jersey 07801.

6.3 Distribution of ammunition data cards. - Distribution of data cards shall include the following: Commander, Picatinny Arsenal, ATTN: SARPA-QA-M Dover, New Jersey 07801.

6.4 Check test for possible deterioration. - The following test is designed to prevent detonators that have deteriorated from being assembled into major ammunition items. The test shall be performed on detonators from each lot which have been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are assembled does not require that the check test be performed, the Contracting Officer should insert in the contract a provision for the performance of the check test. The Contracting Officer will arrange for the contractor to be reimbursed for expense incurred in his performance of the check test for possible deterioration. Detonator assemblies shall be selected from each lot for this test.

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6.4.1 Output. A sample of 315 detonators shall be selected at random from each lot. If 4 or more detonators fail to comply with the requirement specified, the lot shall be rejected. If more than one defective is found, a second sample of 315 shall be selected at random and tested. If in the combined first and second sample a total of 5 or more defectives are found the lot shall be rejected. The test shall be performed as specified in 4.5.5.1 using test equipment as specified in 4.4.4.

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the maringal notations and relationship to the last previous issue.

6.6 Supersession data. This specification includes the requirements of MIL-D-60013A (MU) dated 1 May 1964.

6.7 Delay in Distribution. Document was approved 15 April 1975, however, it was not received by Department of Defense Single Stock Point (DODSSP) for indexing and distribution until January 1980.

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Preparing Activity:
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MIL-D-60014B Detonator, Stab M47, Parts for, LAP

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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